COURSE TITLE : COMPUTER NETWORKS AND NETWORK PROGRAMMING

COURSE CODE : 25CS2208E

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SECTION : 2

SKILL – 1: ENCAPSULATION BUILDER

Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim

To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Introduction

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, real network packets are captured and the Ethernet, IP, and TCP/UDP headers are examined. The total overhead is calculated to understand how efficiently the network carries application data.

Requirements

- Mac computer with an active Wi-Fi or Ethernet connection.
- Wireshark application.
- Internet connection and a web browser.
- Permission to capture packets on the selected network interface.

Part A – Install Wireshark on macOS

1. Open Safari, Chrome, or another browser on your Mac.
2. Visit the official Wireshark download page: https://www.wireshark.org/download.html.
3. Download the macOS installer that matches your Mac.
4. Open the downloaded .dmg file and follow the installer instructions.
5. If the installer asks to install additional capture permissions/components, follow the on-screen instructions. You may need your Mac administrator password.

6. Open Wireshark from Applications. If macOS displays a security confirmation, choose Open only if the application was downloaded from the official Wireshark website.

Screenshot to insert: Wireshark installed and opened on macOS.

[INSERT SCREENSHOT 1 HERE – WIRESHARK APPLICATION OPEN]

Part B – Select the network interface

7. Make sure your Mac is connected to the internet.
8. Open Wireshark. The start screen displays available network interfaces.
9. If you are using wireless internet, select the Wi-Fi interface. If you are connected using a cable, select the relevant Ethernet/USB network interface.
10. Double-click the selected interface to start packet capture.
11. If no packets appear or permission is denied, close Wireshark and check that the capture permissions/components were installed correctly.

Screenshot to insert: Available interfaces and the selected Wi-Fi interface.

[INSERT SCREENSHOT 2 HERE – NETWORK INTERFACE SELECTION]

Part C – Generate network traffic

12. Keep Wireshark capturing packets.
13. Open a new browser tab.
14. Visit https://www.google.com.
15. Open two or three different websites in separate tabs.
16. Wait for approximately 30 seconds so that network packets are generated.
17. Return to Wireshark.
18. Click the red Stop button at the top of the Wireshark window.

You should now see the packets captured from your network activity.

[INSERT SCREENSHOT 3 HERE – CAPTURED PACKET LIST]

Part D – Apply display filters

Enter each filter separately in the display filter bar and press Enter.

Display filter	Purpose
tcp	Displays TCP packets.
udp	Displays UDP packets.
icmp	Displays ICMP packets.

tcp.len > 0	Displays TCP packets carrying application data.

19. Type tcp in the display filter bar and press Enter. Observe the TCP packets.
20. Replace the filter with udp and press Enter. Observe the UDP packets.
21. Replace the filter with icmp and press Enter to view ICMP packets, if any were captured.
22. Enter tcp.len > 0 to view TCP packets that carry data.

[INSERT SCREENSHOT 4 HERE – TCP FILTER]

[INSERT SCREENSHOT 5 HERE – UDP FILTER]

Part E – Inspect a TCP packet header

23. Apply the tcp display filter.
24. Select any TCP packet from the packet list.
25. In the packet details pane, expand Frame.
26. Expand Ethernet II and observe the source and destination MAC addresses.
27. Expand Internet Protocol Version 4 and observe the IPv4 header length and total length.
28. Expand Transmission Control Protocol and observe the TCP header length and segment information.
29. Record the values in the observation table provided in this record.

Parameter	**Where to find it**
Frame length	Frame
Ethernet header and source/destination MAC addresses	Ethernet II
IPv4 header length and IPv4 total length	Internet Protocol Version 4
TCP header length and TCP segment length	Transmission Control Protocol

[INSERT SCREENSHOT 6 HERE – EXPANDED TCP PACKET DETAILS]

Part F – Inspect a UDP packet header

30. Replace the display filter with udp and press Enter.
31. Select any UDP packet from the packet list.
32. Expand Frame, Ethernet II, Internet Protocol Version 4, and User Datagram Protocol in the packet details pane.
33. Observe the UDP header and its Length field.

34. Record the values in the observation table.

[INSERT SCREENSHOT 7 HERE – EXPANDED UDP PACKET DETAILS]

Part G – Calculate header overhead

For the basic example given in the lab sheet, the packet contains the following:

Component	Size
Ethernet header	14 bytes
IPv4 header	20 bytes
TCP header	20 bytes
Application data	1500 bytes
FCS (if included)	4 bytes

1. Total size including FCS

Total size = Ethernet header + IPv4 header + TCP header + Application data + FCS

= 14 + 20 + 20 + 1500 + 4

= 1558 bytes

2. Total overhead

Overhead = Ethernet header + IPv4 header + TCP header + FCS

= 14 + 20 + 20 + 4

= 58 bytes

3. Payload efficiency

Payload efficiency = (Application data ÷ Total size) × 100

= (1500 ÷ 1558) × 100

≈ 96.28%

Note: The lab sheet says Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording observations. The 1558-byte calculation is the lab sheet's illustrative arithmetic example; actual captured packet sizes can differ, and Ethernet MTU/frame-size limits may mean that a 1500-byte application-data example is not represented as one ordinary Ethernet frame.

Part I – Save the packet capture

35. Start packet capture in Wireshark by double-clicking your Wi-Fi or Ethernet interface.

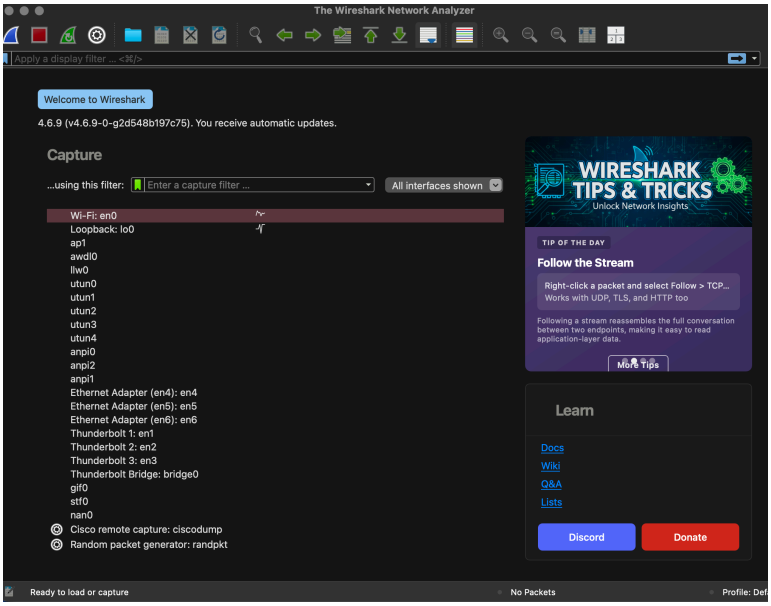
36. Generate sufficient network traffic by visiting a few websites.
37. Click the red Stop button to stop the capture.
38. Choose File → Save As.
39. Select the folder where you want to save the capture.
40. Enter a filename such as sender_capture.pcapng.
41. Keep the file format as pcapng and click Save.

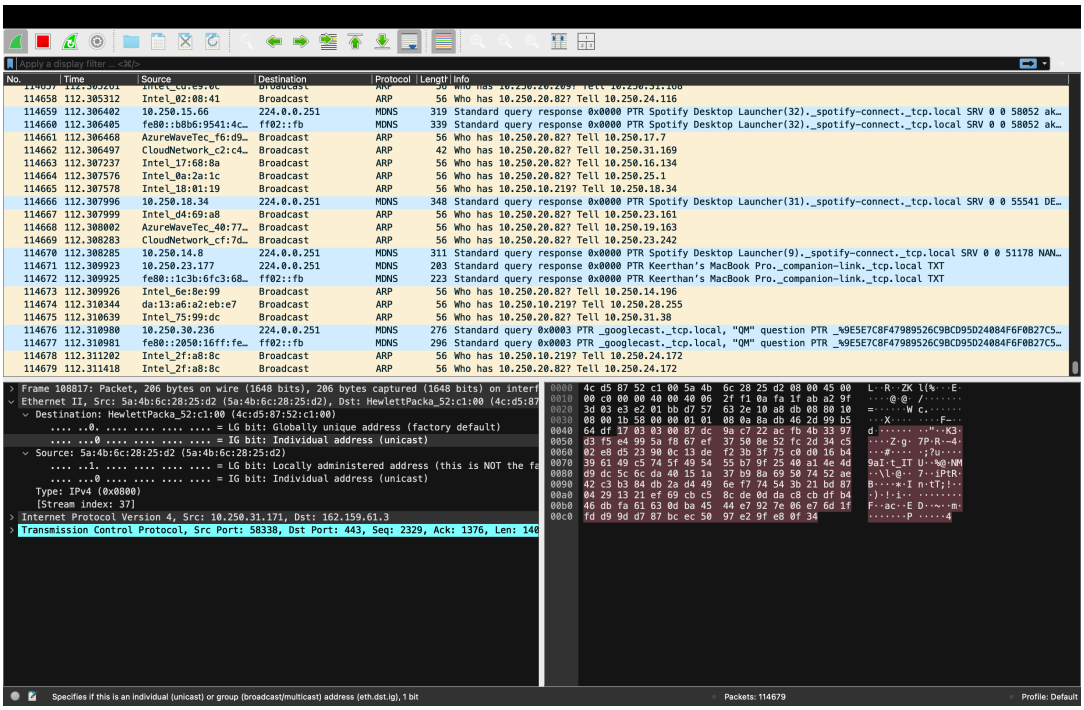
[INSERT SCREENSHOT 8 HERE – SAVE AS WINDOW OR SAVED CAPTURE]

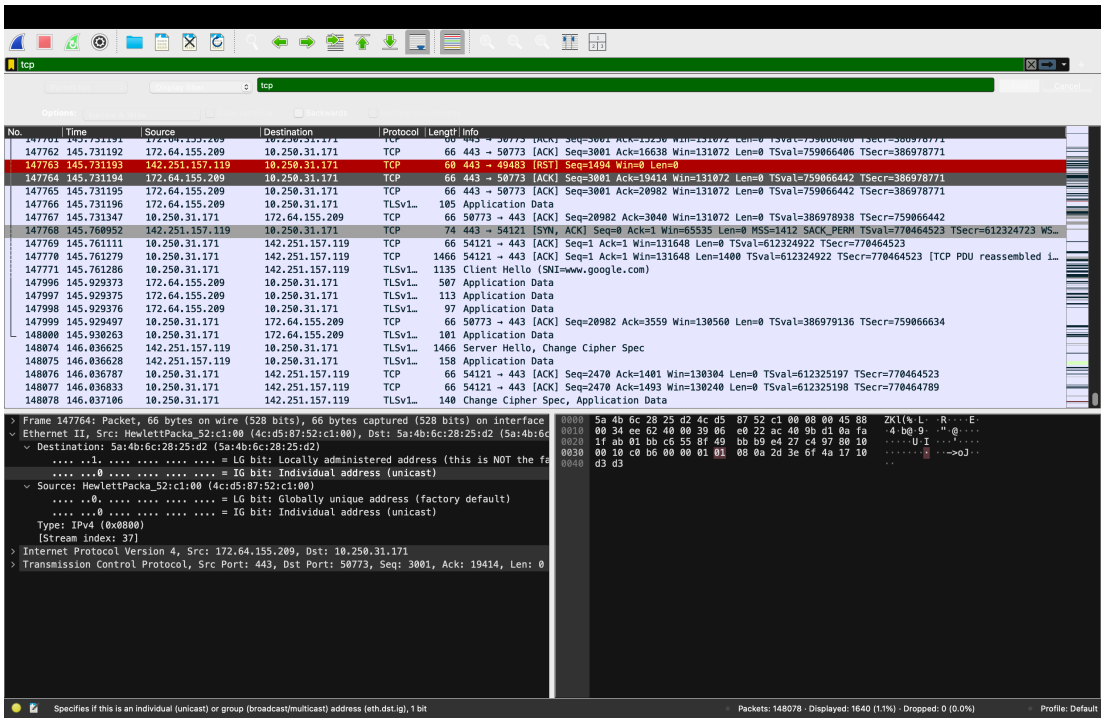
Part J – Open a saved capture file

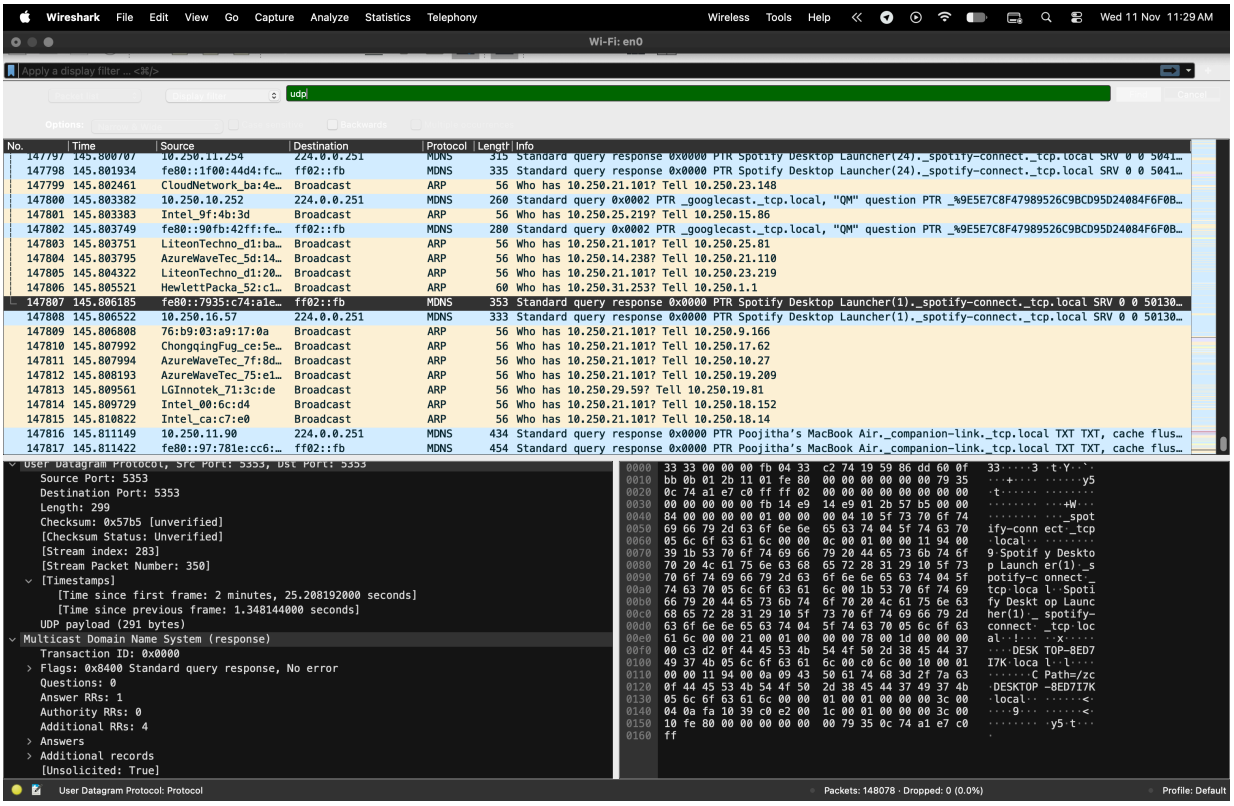
42. Open Wireshark.
43. Choose File → Open.
44. Select the saved .pcapng or .pcap file.
45. Click Open.
46. Enter icmp in the display filter bar and press Enter to view ICMP packets, if present in the saved capture.

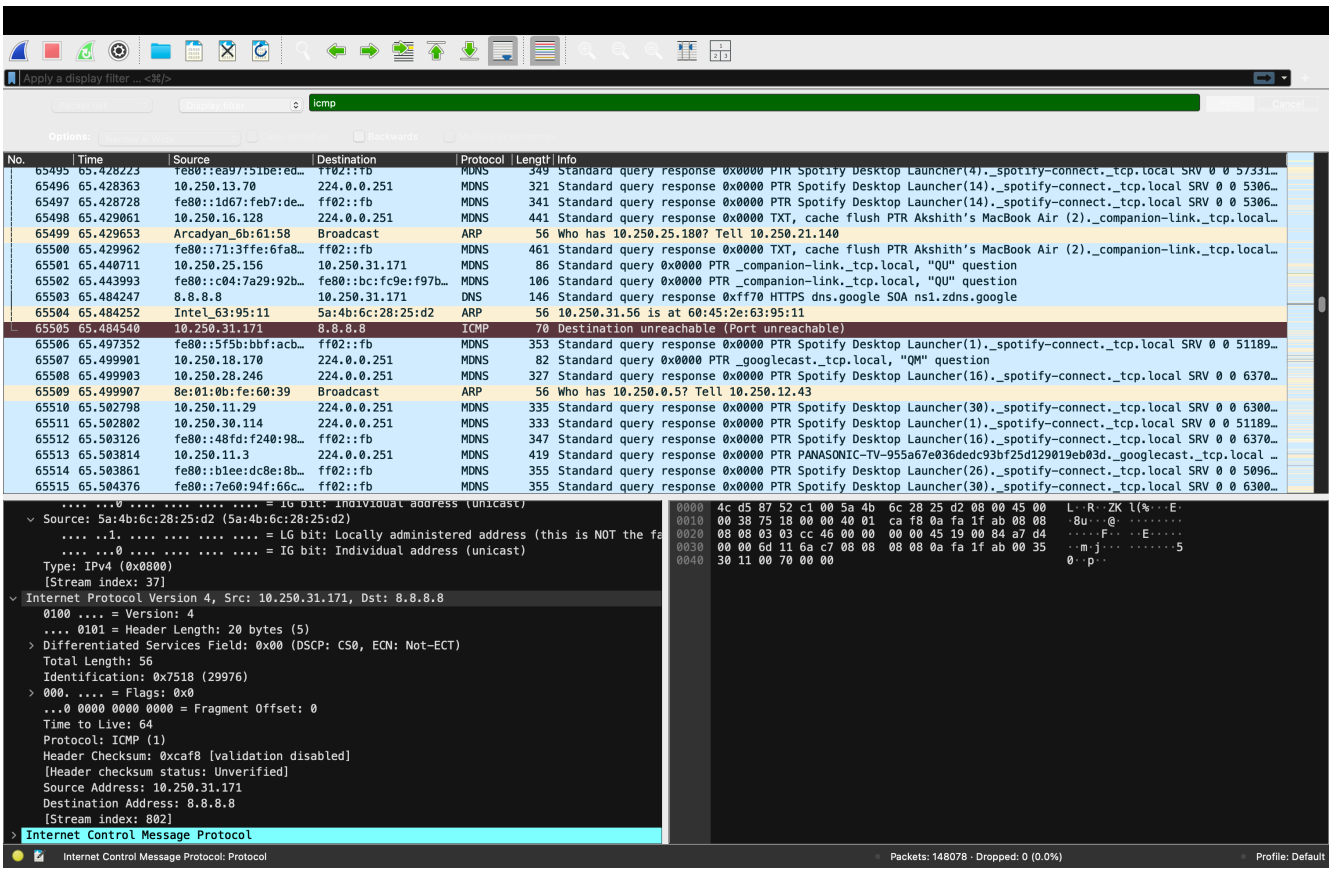
PROCESS SCREENSHOTS AND OUTPUTS

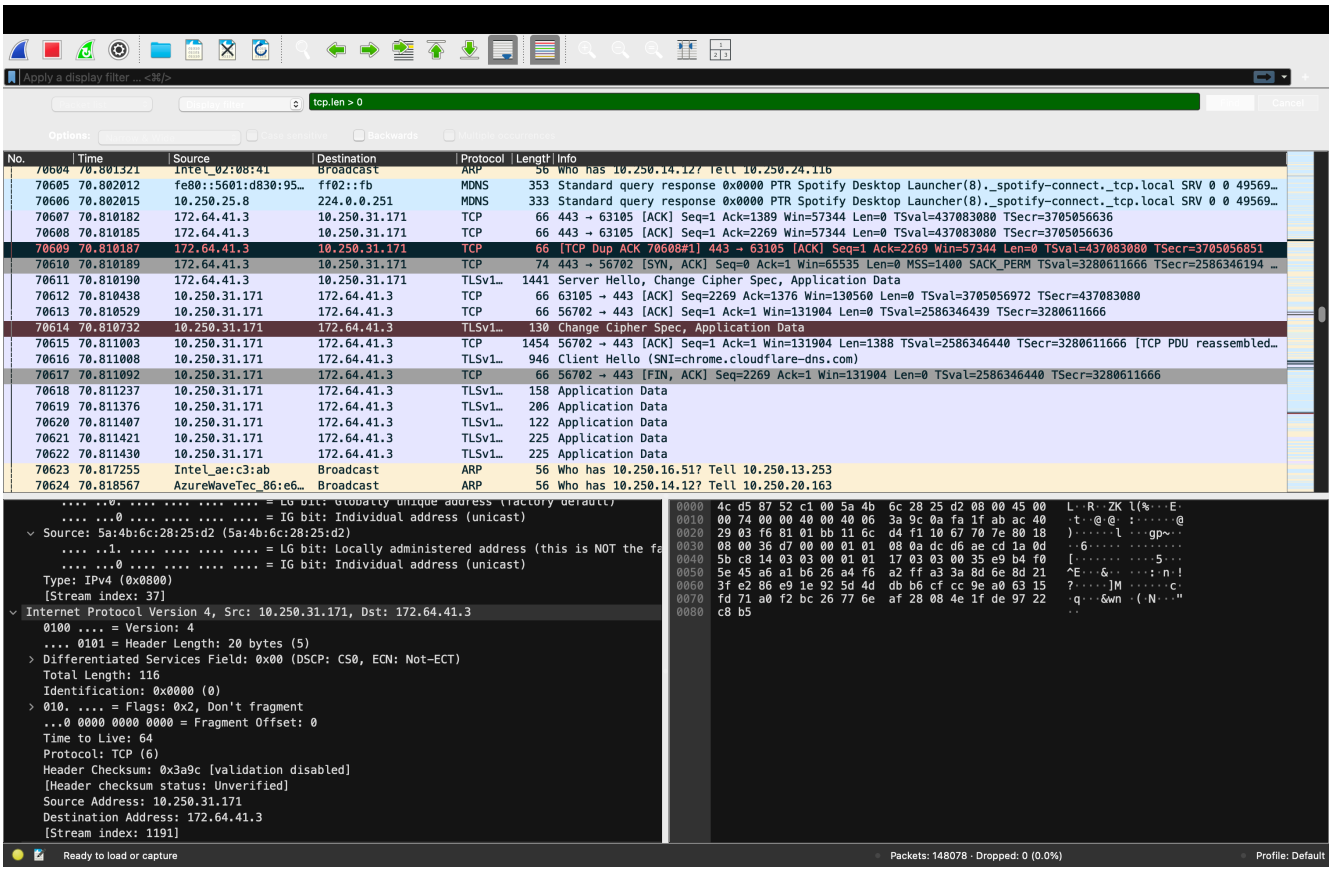




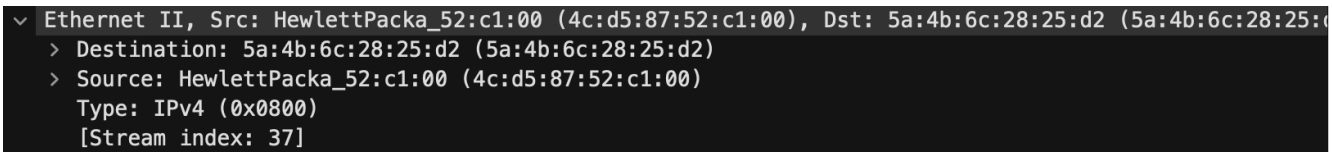


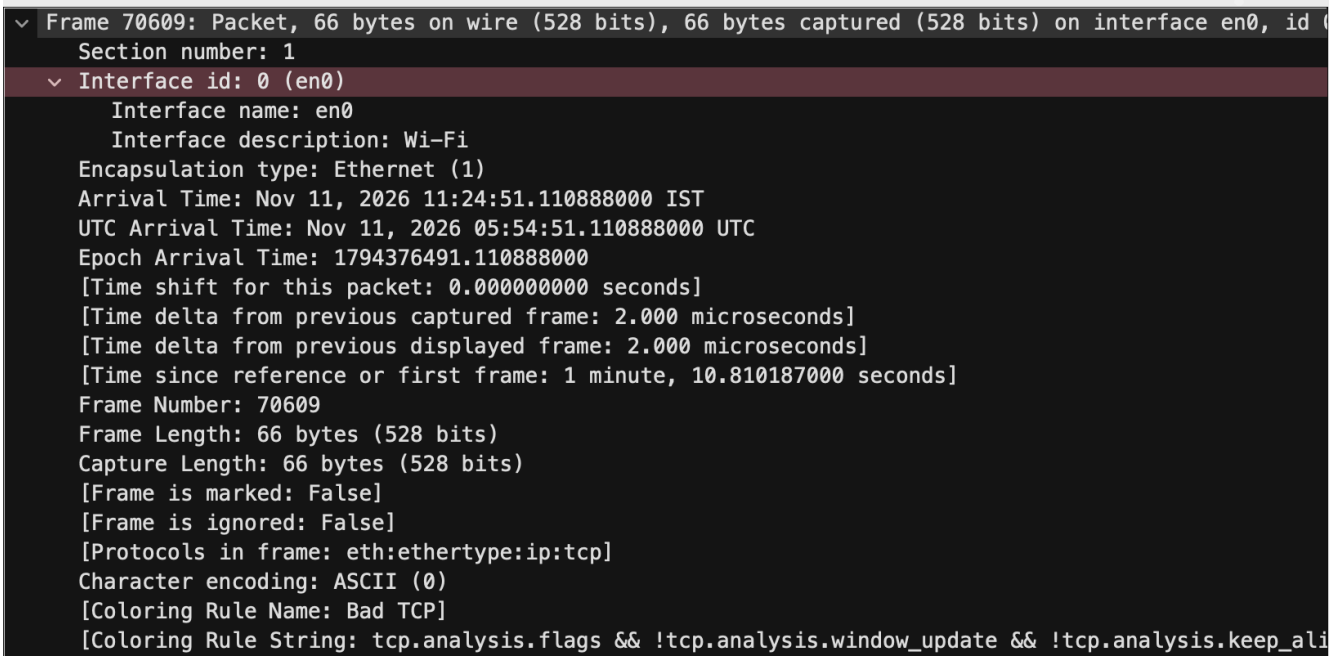

Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help Wed 11 Nov 11:29 AM
Wi-Fi: en0
udp
No. Time Source Destination Protocol Length Info
147798 145.801934 fe80::1f00:44d4:fc… ff02::fb MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(24)._spotify-connect._tcp.local SRV 0 0 5041…
147799 145.802461 CloudNetwork_ba:4e… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.23.148
147800 145.803382 10.250.10.252 224.0.0.251 MDNS 260 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B…
147801 145.803383 Intel_9f:4b:3d Broadcast ARP 56 Who has 10.250.25.219? Tell 10.250.15.86
147802 145.803749 fe80::90fb:42ff:fe… ff02::fb MDNS 280 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B…
147803 145.803751 LiteonTechno_d1:ba… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.25.81
147804 145.803795 AzureWaveTec_5d:14… Broadcast ARP 56 Who has 10.250.14.238? Tell 10.250.21.110
147805 145.804322 LiteonTechno_d1:20… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.23.219
147806 145.805521 HewlettPacka_52:c1… Broadcast ARP 60 Who has 10.250.31.253? Tell 10.250.1.1
147807 145.806185 fe80::7935:c74:a1e… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 50130…
147808 145.806522 10.250.16.57 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 50130…
147809 145.806808 76:b9:03:a9:17:0a Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.9.166
147810 145.807992 ChongqingFug_ce:5e… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.17.62
147811 145.807994 AzureWaveTec_7f:8d… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.10.27
147812 145.808193 AzureWaveTec_75:e1… Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.19.209
147813 145.809561 LGInnotek_71:3c:de Broadcast ARP 56 Who has 10.250.29.59? Tell 10.250.19.81
147814 145.809729 Intel_00:6c:d4 Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.18.152
147815 145.810822 Intel_ca:c7:e0 Broadcast ARP 56 Who has 10.250.21.101? Tell 10.250.18.14
147816 145.811149 10.250.11.90 224.0.0.251 MDNS 434 Standard query response 0x0000 PTR Poojitha's MacBook Air._companion-link._tcp.local TXT TXT, cache flus…
147817 145.811422 fe80::97:781e:cc6:… ff02::fb MDNS 454 Standard query response 0x0000 PTR Poojitha's MacBook Air._companion-link._tcp.local TXT TXT, cache flus…
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
Destination Port: 5353
Length: 299
Checksum: 0x57b5 [unverified]
[Checksum Status: Unverified]
[Stream index: 283]
[Stream Packet Number: 350]
[Timestamps]
[Time since first frame: 2 minutes, 25.208192000 seconds]
[Time since previous frame: 1.348144000 seconds]
UDP payload (291 bytes)
Multicast Domain Name System (response)
Transaction ID: 0x0000
Flags: 0x8400 Standard query response, No error
Questions: 0
Answer RRs: 1
Authority RRs: 0
Additional RRs: 4
Answers
Additional records
[Unsolicited: True]
0000 33 33 00 00 00 fb 04 33 c2 74 19 59 86 dd 60 0f
0010 bb 0b 01 2b 11 01 fe 80 00 00 00 00 00 00 79 35
0020 0c 74 a1 e7 c0 ff ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 fb 14 e9 14 e9 01 2b 57 b5 00 00
0040 84 00 00 00 00 01 00 00 00 04 10 5f 73 70 6f 74
0050 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70
0060 05 6c 6f 63 61 6c 00 00 0c 00 01 00 00 11 94 00
0070 39 1b 53 70 6f 74 69 66 79 20 44 65 73 6b 74 6f
0080 70 20 4c 61 75 6e 63 68 65 72 28 31 29 10 5f 73
0090 70 6f 74 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f
00a0 74 63 70 05 6c 6f 63 61 6c 00 1b 53 70 6f 74 69
00b0 66 79 20 44 65 73 6b 74 6f 70 20 4c 61 75 6e 63
00c0 68 65 72 28 31 29 10 5f 73 70 6f 74 69 66 79 2d
00d0 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63
00e0 61 6c 00 00 21 00 01 00 00 00 78 00 1d 00 00 00
00f0 00 c3 d2 0f 44 45 53 4b 54 4f 50 2d 38 45 44 37
0100 49 37 4b 05 6c 6f 63 61 6c 00 c0 6c 00 10 00 01
0110 00 00 11 94 00 0a 09 43 50 61 74 68 3d 2f 7a 63
0120 0f 44 45 53 4b 54 4f 50 2d 38 45 44 37 49 37 4b
0130 05 6c 6f 63 61 6c 00 00 01 00 01 00 00 00 3c 00
0140 04 0a fa 10 39 c0 e2 00 1c 00 01 00 00 00 3c 00
0150 10 fe 80 00 00 00 00 00 00 79 35 0c 74 a1 e7 c0
0160 ff
User Datagram Protocol: Protocol
Packets: 148078 · Dropped: 0 (0.0%)
Profile: Default


icmp
No. Time Source Destination Protocol Length Info
65496 65.428363 10.250.13.70 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR Spotify Desktop Launcher(14)._spotify-connect._tcp.local SRV 0 0 5306…
65497 65.428728 fe80::1d67:feb7:de… ff02::fb MDNS 341 Standard query response 0x0000 PTR Spotify Desktop Launcher(14)._spotify-connect._tcp.local SRV 0 0 5306…
65498 65.429061 10.250.16.128 224.0.0.251 MDNS 441 Standard query response 0x0000 TXT, cache flush PTR Akshith's MacBook Air (2)._companion-link._tcp.local…
65499 65.429653 Arcadyan_6b:61:58 Broadcast ARP 56 Who has 10.250.25.180? Tell 10.250.21.140
65500 65.429962 fe80::71:3ffe:6fa8… ff02::fb MDNS 461 Standard query response 0x0000 TXT, cache flush PTR Akshith's MacBook Air (2)._companion-link._tcp.local…
65501 65.440711 10.250.25.156 10.250.31.171 MDNS 86 Standard query 0x0000 PTR _companion-link._tcp.local, "QU" question
65502 65.443993 fe80::c04:7a29:92b… fe80::bc:fc9e:f97b… MDNS 106 Standard query 0x0000 PTR _companion-link._tcp.local, "QU" question
65503 65.484247 8.8.8.8 10.250.31.171 DNS 146 Standard query response 0xff70 HTTPS dns.google SOA ns1.zdns.google
65504 65.484252 Intel_63:95:11 5a:4b:6c:28:25:d2 ARP 56 10.250.31.56 is at 60:45:2e:63:95:11
65505 65.484540 10.250.31.171 8.8.8.8 ICMP 70 Destination unreachable (Port unreachable)
65506 65.497352 fe80::5f5b:bbf:acb… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 51189…
65507 65.499901 10.250.18.170 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
65508 65.499903 10.250.28.246 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher(16)._spotify-connect._tcp.local SRV 0 0 6370…
65509 65.499907 8e:01:0b:fe:60:39 Broadcast ARP 56 Who has 10.250.0.5? Tell 10.250.12.43
65510 65.502798 10.250.11.29 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(30)._spotify-connect._tcp.local SRV 0 0 6300…
65511 65.502802 10.250.30.114 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 51189…
65512 65.503126 fe80::48fd:f240:98… ff02::fb MDNS 347 Standard query response 0x0000 PTR Spotify Desktop Launcher(16)._spotify-connect._tcp.local SRV 0 0 6370…
65513 65.503814 10.250.11.3 224.0.0.251 MDNS 419 Standard query response 0x0000 PTR PANASONIC-TV-955a67e036dedc93bf25d129019eb03d._googlecast._tcp.local …
65514 65.503861 fe80::b1ee:dc8e:8b… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(26)._spotify-connect._tcp.local SRV 0 0 5096…
65515 65.504376 fe80::7e60:94f:66c… ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(30)._spotify-connect._tcp.local SRV 0 0 6300…
Source: 5a:4b:6c:28:25:d2 (5a:4b:6c:28:25:d2)
Type: IPv4 (0x0800)
[Stream index: 37]
Internet Protocol Version 4, Src: 10.250.31.171, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0x7518 (29976)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0xcaf8 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.31.171
Destination Address: 8.8.8.8
[Stream index: 802]
Internet Control Message Protocol
0000 4c d5 87 52 c1 00 5a 4b 6c 28 25 d2 08 00 45 00
0010 00 38 75 18 00 00 40 01 ca f8 0a fa 1f ab 08 08
0020 08 08 03 03 cc 46 00 00 00 00 45 19 00 84 a7 d4
0030 00 00 6d 11 6a c7 08 08 08 08 0a fa 1f ab 00 35
0040 30 11 00 70 00 00
Internet Control Message Protocol: Protocol
Packets: 148078 · Dropped: 0 (0.0%)
Profile: Default

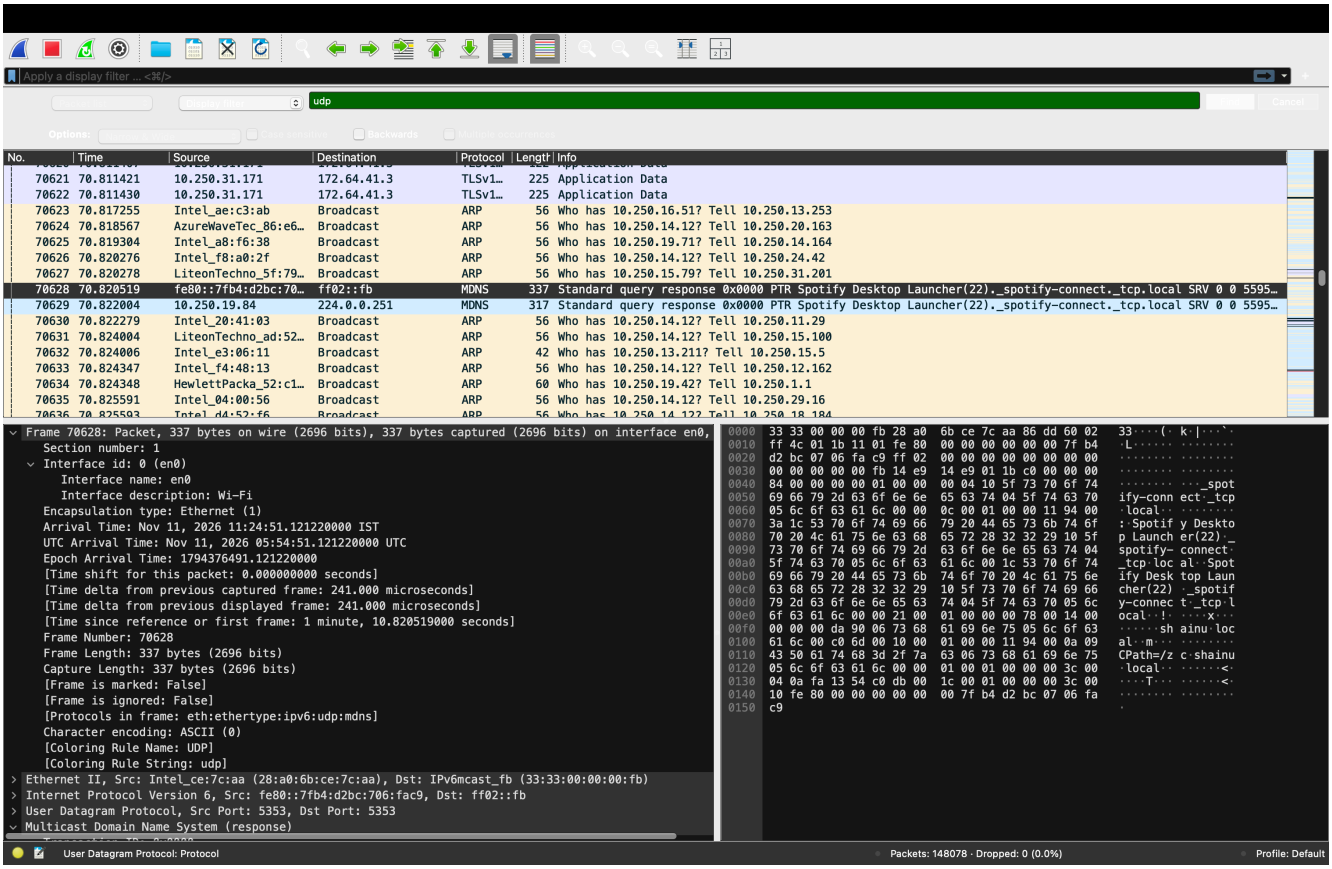


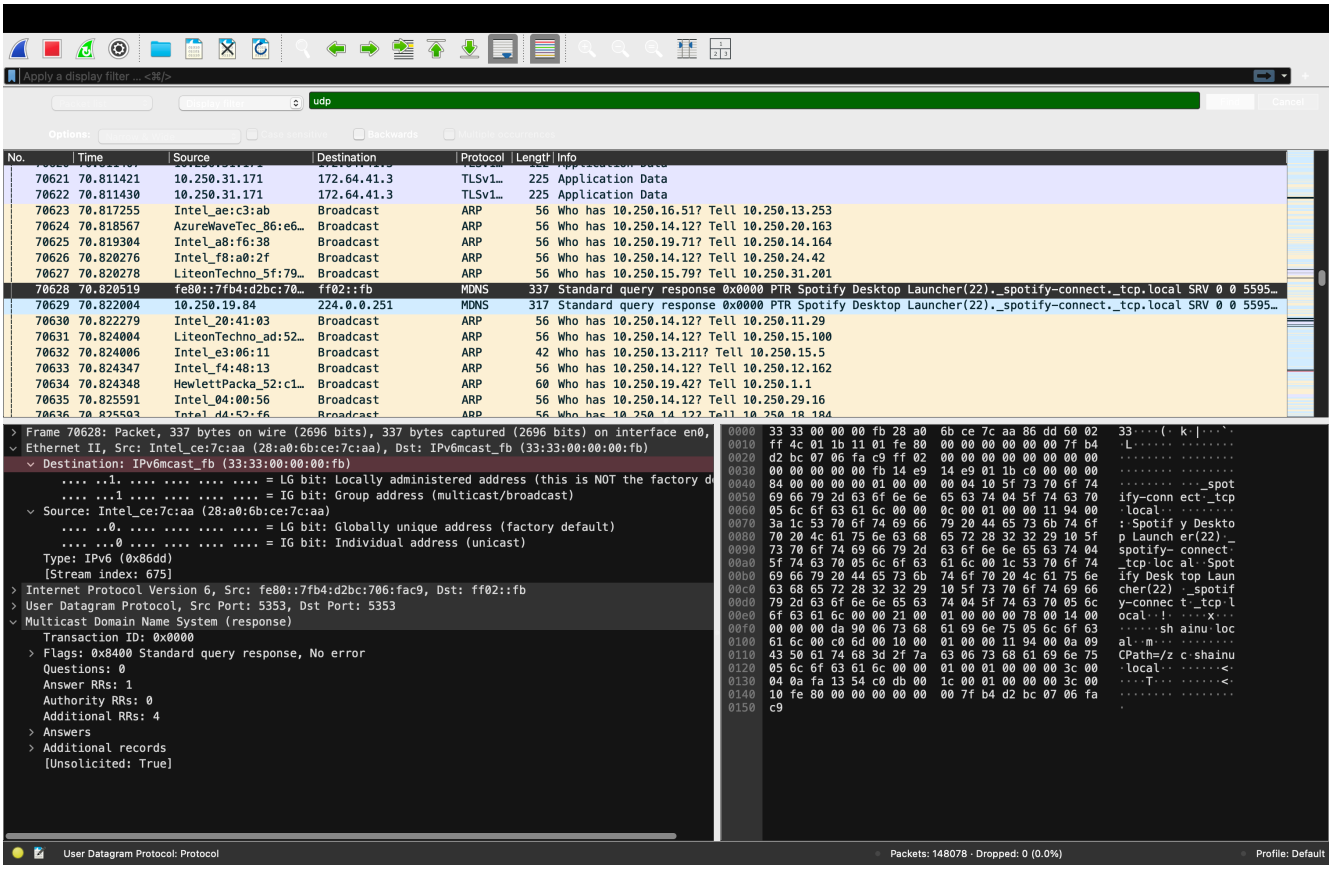

```
Internet Protocol Version 4, Src: 172.64.41.3, Dst: 10.250.31.171
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x88 (DSCP: AF41, ECN: Not-ECT)
    Total Length: 52
    Identification: 0xd1ce (53710)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 63
    Protocol: TCP (6)
    Header Checksum: 0xa985 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.64.41.3
    Destination Address: 10.250.31.171
    [Stream index: 1191]
```

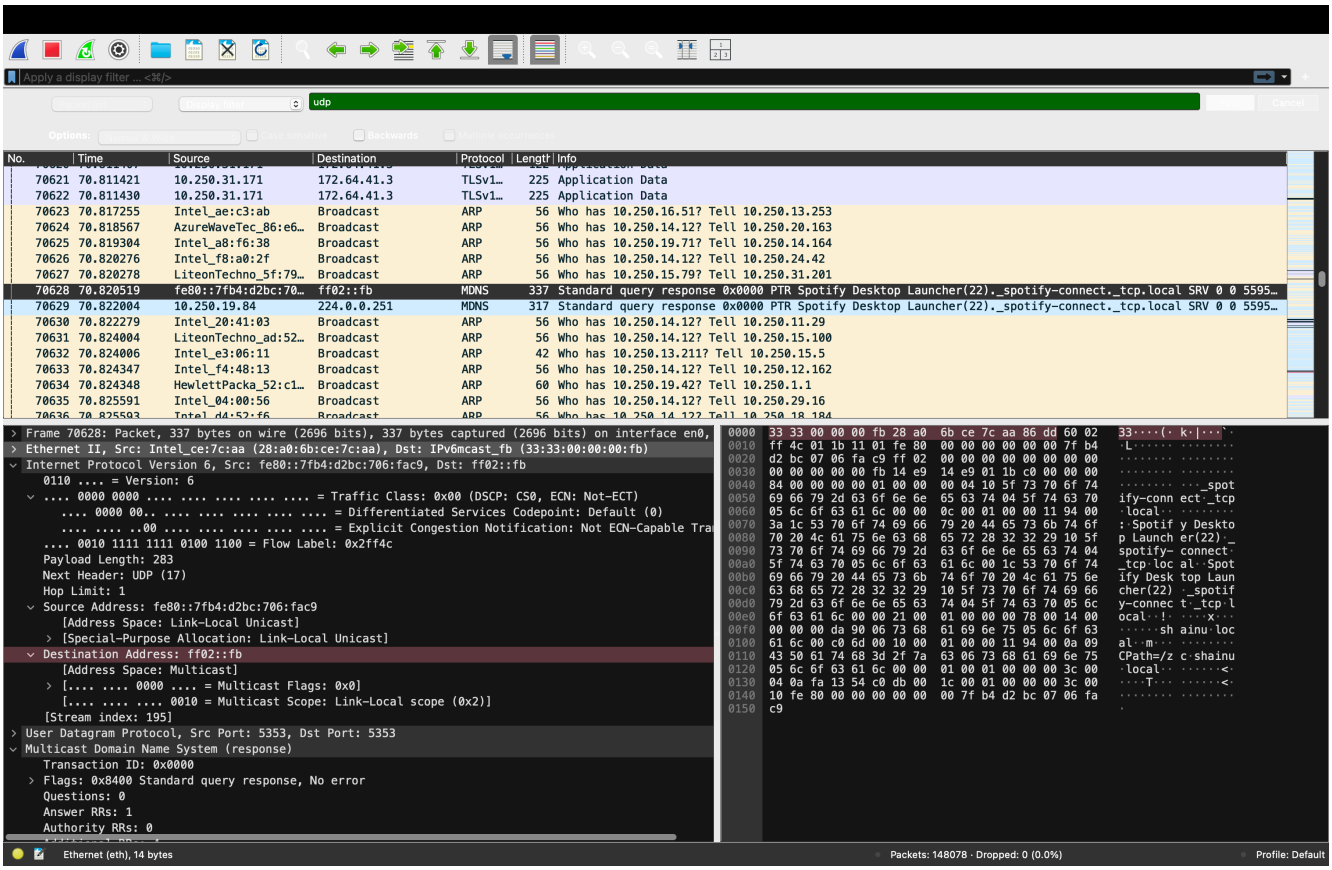

Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: 5a:4b:6c:28:25:d2 (5a:4b:6c:28:25:
Destination: 5a:4b:6c:28:25:d2 (5a:4b:6c:28:25:d2)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Type: IPv4 (0x0800)
[Stream index: 37]

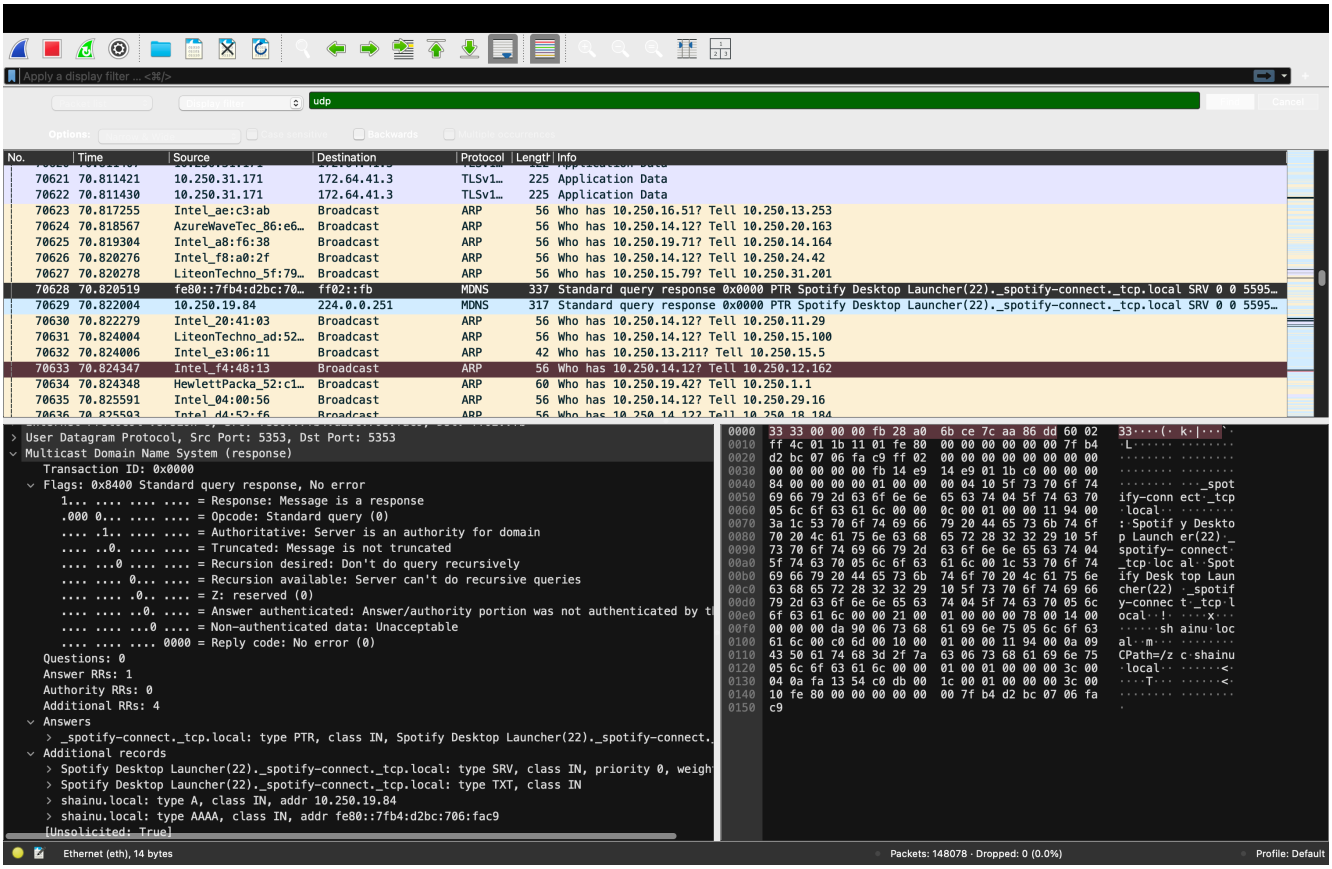

Frame 70609: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface en0, id
Section number: 1
Interface id: 0 (en0)
Interface name: en0
Interface description: Wi-Fi
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 11:24:51.110888000 IST
UTC Arrival Time: Nov 11, 2026 05:54:51.110888000 UTC
Epoch Arrival Time: 1794376491.110888000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.000 microseconds]
[Time delta from previous displayed frame: 2.000 microseconds]
[Time since reference or first frame: 1 minute, 10.810187000 seconds]
Frame Number: 70609
Frame Length: 66 bytes (528 bits)
Capture Length: 66 bytes (528 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: Bad TCP]
[Coloring Rule String: tcp.analysis.flags && !tcp.analysis.window_update && !tcp.analysis.keep_ali

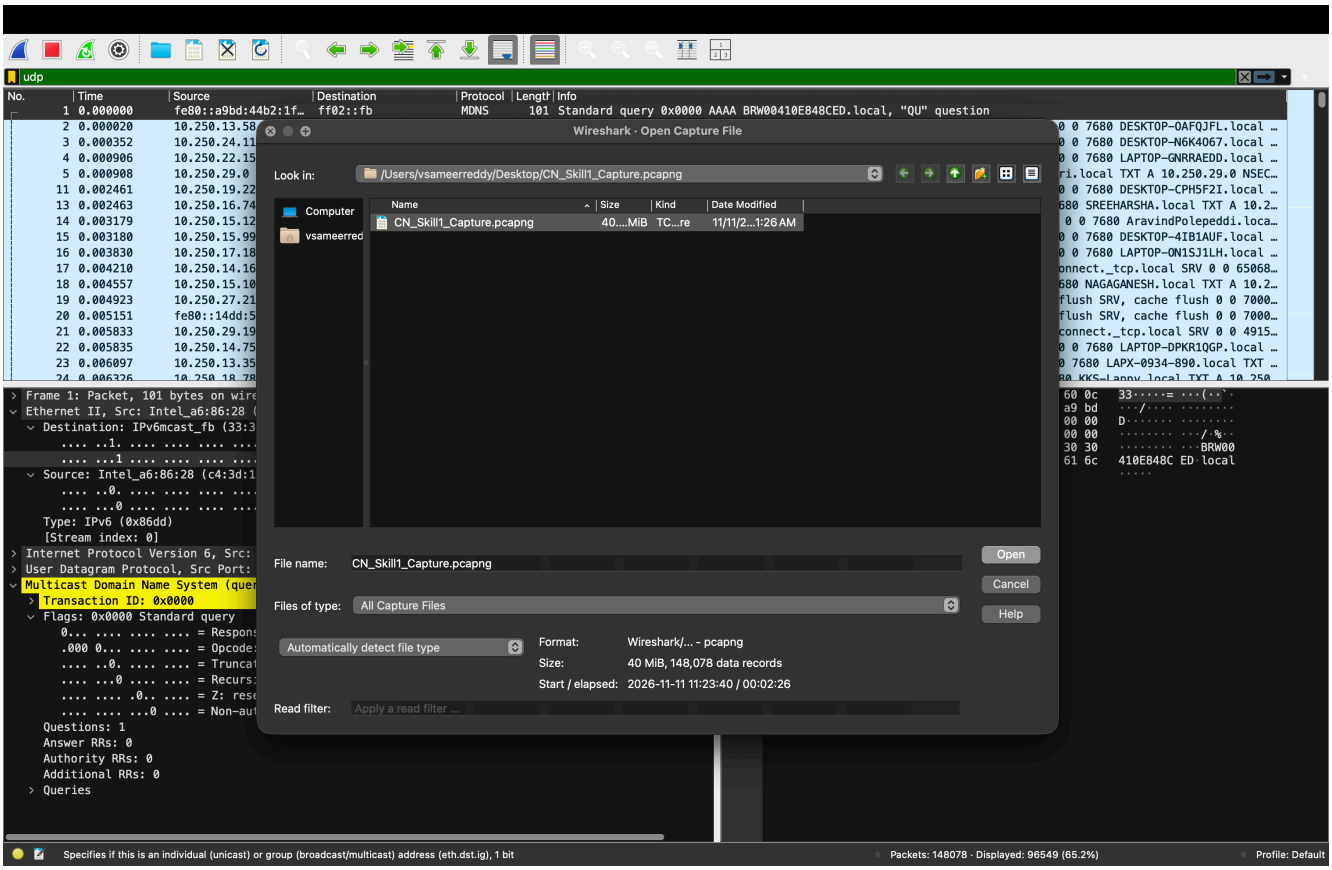
```
Transmission Control Protocol, Src Port: 443, Dst Port: 63105, Seq: 1, Ack: 2269, Len: 0
    Source Port: 443
    Destination Port: 63105
    [Stream index: 48]
    [Stream Packet Number: 9]
  > [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 275213087
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 2269    (relative ack number)
    Acknowledgment number (raw): 292345073
    1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x010 (ACK)
    Window: 7
    [Calculated window size: 57344]
    [Window size scaling factor: 8192]
    Checksum: 0x1bad [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

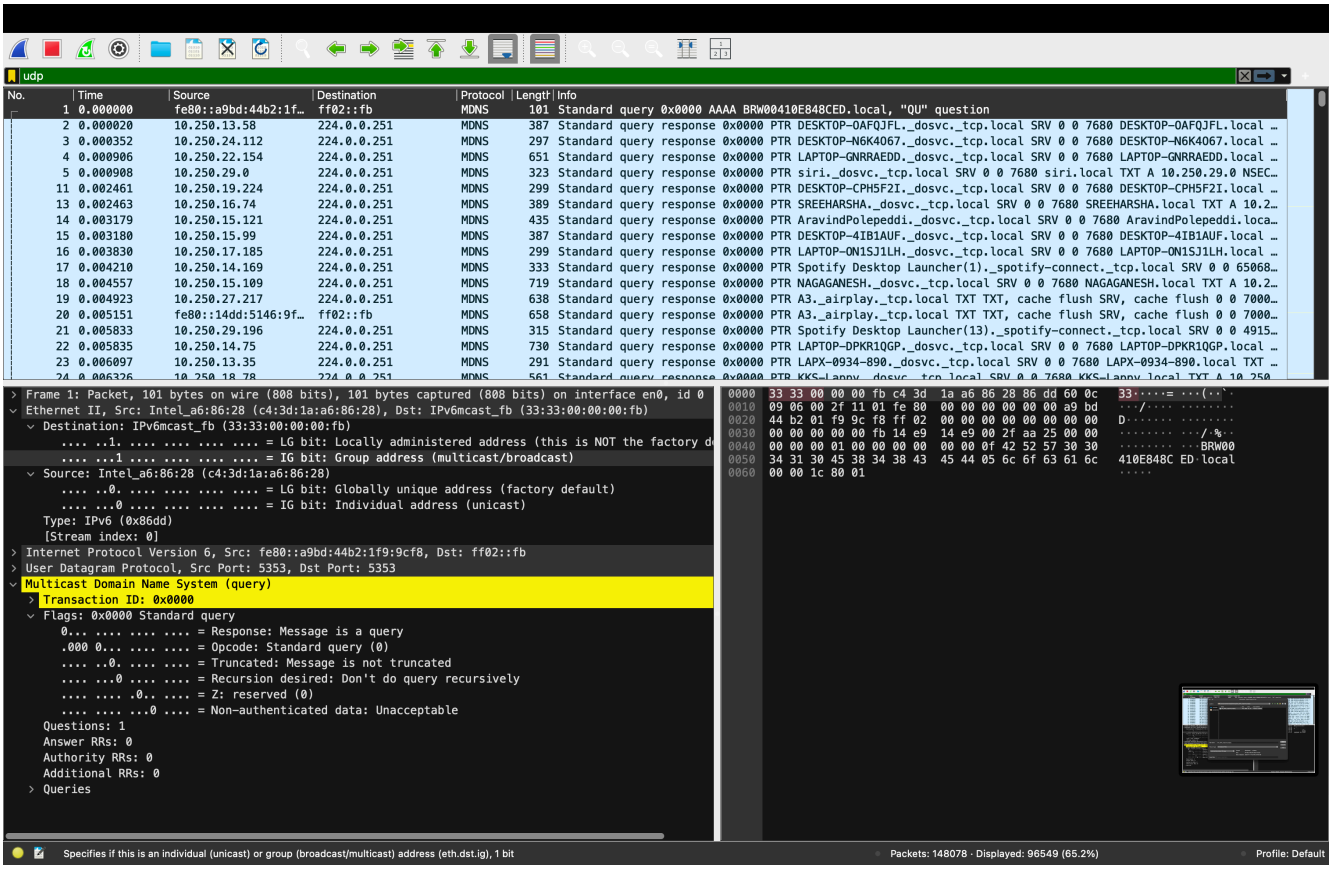


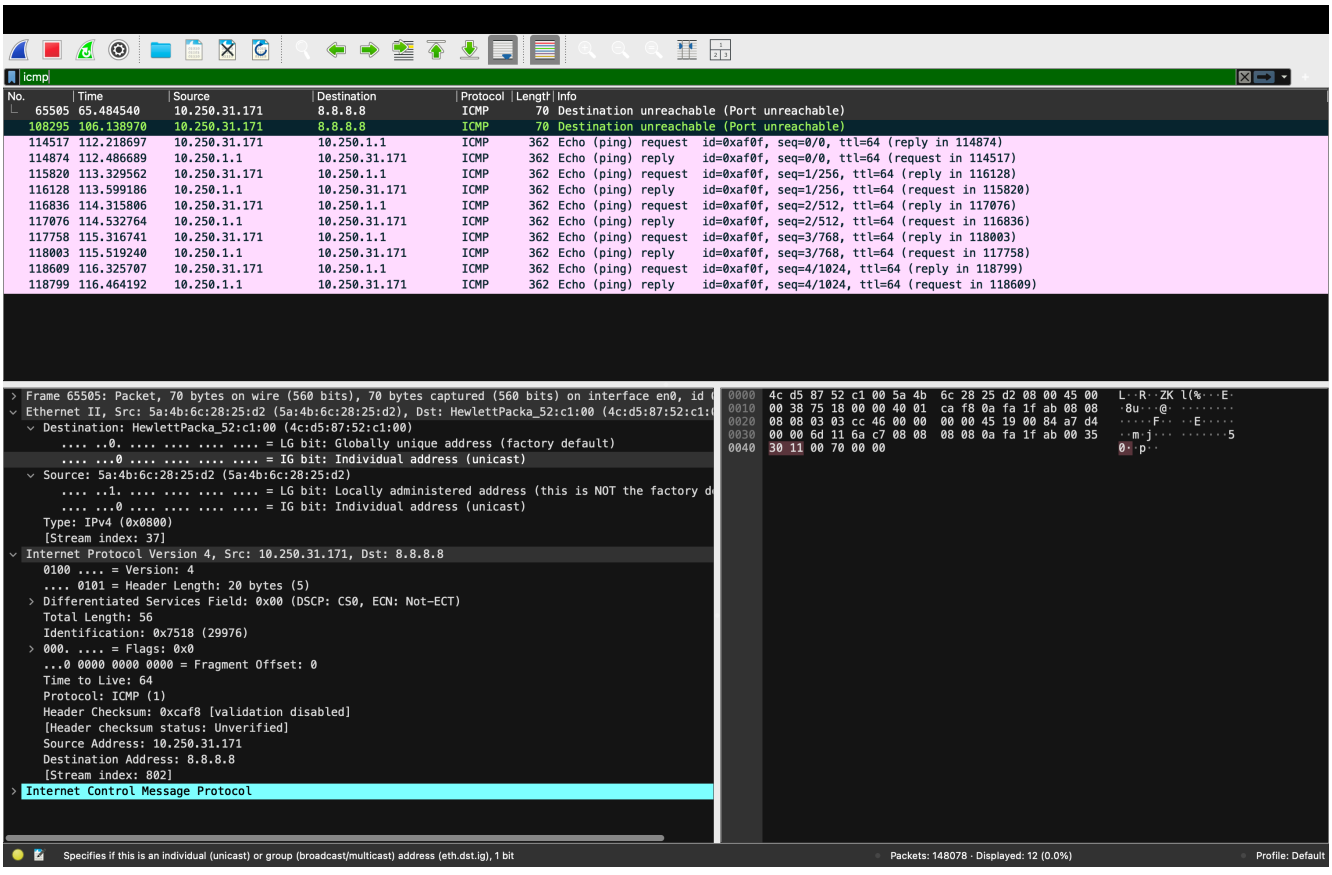

udp
No. Time Source Destination Protocol Lengt Info
70621 70.811421 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70622 70.811430 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70623 70.817255 Intel_ae:c3:ab Broadcast ARP 56 Who has 10.250.16.51? Tell 10.250.13.253
70624 70.818567 AzureWaveTec_86:e6… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.20.163
70625 70.819304 Intel_a8:f6:38 Broadcast ARP 56 Who has 10.250.19.71? Tell 10.250.14.164
70626 70.820276 Intel_f8:a0:2f Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.24.42
70627 70.820278 LiteonTechno_5f:79… Broadcast ARP 56 Who has 10.250.15.79? Tell 10.250.31.201
70628 70.820519 fe80::7fb4:d2bc:70… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70629 70.822004 10.250.19.84 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70630 70.822279 Intel_20:41:03 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.11.29
70631 70.824004 LiteonTechno_ad:52… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.15.100
70632 70.824006 Intel_e3:06:11 Broadcast ARP 42 Who has 10.250.13.211? Tell 10.250.15.5
70633 70.824347 Intel_f4:48:13 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.12.162
70634 70.824348 HewlettPacka_52:c1… Broadcast ARP 60 Who has 10.250.19.42? Tell 10.250.1.1
70635 70.825591 Intel_04:00:56 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.29.16
Frame 70628: Packet, 337 bytes on wire (2696 bits), 337 bytes captured (2696 bits) on interface en0,
Ethernet II, Src: Intel_ce:7c:aa (28:a0:6b:ce:7c:aa), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
Destination: IPv6mcast_fb (33:33:00:00:00:fb)
.... ..1. .... = LG bit: Locally administered address (this is NOT the factory d
....1 .... = IG bit: Group address (multicast/broadcast)
Source: Intel_ce:7c:aa (28:a0:6b:ce:7c:aa)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv6 (0x86dd)
[Stream index: 675]
Internet Protocol Version 6, Src: fe80::7fb4:d2bc:706:fac9, Dst: ff02::fb
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
Transaction ID: 0x0000
Flags: 0x8400 Standard query response, No error
Questions: 0
Answer RRs: 1
Authority RRs: 0
Additional RRs: 4
Answers
Additional records
[Unsolicited: True]
0000 33 33 00 00 00 fb 28 a0 6b ce 7c aa 86 dd 60 02 33····(· k·|···`
0010 ff 4c 01 1b 11 01 fe 80 00 00 00 00 00 00 7f b4 ·L·····
0020 d2 bc 07 06 fa c9 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 fb 14 e9 14 e9 01 1b c0 00 00 00
0040 84 00 00 00 00 01 00 00 00 04 10 5f 73 70 6f 74 _spot
0050 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70 ify-conn ect·_tcp
0060 05 6c 6f 63 61 6c 00 00 0c 00 01 00 00 11 94 00 ·local··
0070 3a 1c 53 70 6f 74 69 66 79 20 44 65 73 6b 74 6f :·Spotif y Deskto
0080 70 20 4c 61 75 6e 63 68 65 72 28 32 32 29 10 5f p Launch er(22)·_
0090 73 70 6f 74 69 66 79 2d 63 6f 6e 6e 65 63 74 04 spotify- connect·
00a0 5f 74 63 70 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 _tcp·loc al··Spot
00b0 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61 75 6e ify Desk top Laun
00c0 63 68 65 72 28 32 32 29 10 5f 73 70 6f 74 69 66 cher(22) ·_spotif
00d0 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c y-connec t·_tcp·l
00e0 6f 63 61 6c 00 00 21 00 01 00 00 00 78 00 14 00 ocal··!· ····x···
00f0 00 00 00 da 90 06 73 68 61 69 6e 75 05 6c 6f 63 ·····sh ainu·loc
0100 61 6c 00 c0 6d 00 10 00 01 00 00 11 94 00 0a 09 al··m···
0110 43 50 61 74 68 3d 2f 7a 63 06 73 68 61 69 6e 75 CPath=/z c·shainu
0120 05 6c 6f 63 61 6c 00 00 01 00 01 00 00 00 3c 00 ·local·· ·····<·
0130 04 0a fa 13 54 c0 db 00 1c 00 01 00 00 00 3c 00 ····T··· ·····<·
0140 10 fe 80 00 00 00 00 00 00 7f b4 d2 bc 07 06 fa
0150 c9
User Datagram Protocol: Protocol
Packets: 148078 · Dropped: 0 (0.0%)
Profile: Default


udp
No. Time Source Destination Protocol Lengt Info
70621 70.811421 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70622 70.811430 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70623 70.817255 Intel_ae:c3:ab Broadcast ARP 56 Who has 10.250.16.51? Tell 10.250.13.253
70624 70.818567 AzureWaveTec_86:e6… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.20.163
70625 70.819304 Intel_a8:f6:38 Broadcast ARP 56 Who has 10.250.19.71? Tell 10.250.14.164
70626 70.820276 Intel_f8:a0:2f Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.24.42
70627 70.820278 LiteonTechno_5f:79… Broadcast ARP 56 Who has 10.250.15.79? Tell 10.250.31.201
70628 70.820519 fe80::7fb4:d2bc:70… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70629 70.822004 10.250.19.84 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70630 70.822279 Intel_20:41:03 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.11.29
70631 70.824004 LiteonTechno_ad:52… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.15.100
70632 70.824006 Intel_e3:06:11 Broadcast ARP 42 Who has 10.250.13.211? Tell 10.250.15.5
70633 70.824347 Intel_f4:48:13 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.12.162
70634 70.824348 HewlettPacka_52:c1… Broadcast ARP 60 Who has 10.250.19.42? Tell 10.250.1.1
70635 70.825591 Intel_04:00:56 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.29.16
Frame 70628: Packet, 337 bytes on wire (2696 bits), 337 bytes captured (2696 bits) on interface en0,
Ethernet II, Src: Intel_ce:7c:aa (28:a0:6b:ce:7c:aa), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
Internet Protocol Version 6, Src: fe80::7fb4:d2bc:706:fac9, Dst: ff02::fb
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
....00 .... = Explicit Congestion Notification: Not ECN-Capable Tra
.... 0010 1111 1111 0100 1100 = Flow Label: 0x2ff4c
Payload Length: 283
Next Header: UDP (17)
Hop Limit: 1
Source Address: fe80::7fb4:d2bc:706:fac9
[Address Space: Link-Local Unicast]
[Special-Purpose Allocation: Link-Local Unicast]
Destination Address: ff02::fb
[Address Space: Multicast]
[.... .... 0000 .... = Multicast Flags: 0x0]
[.... .... 0010 = Multicast Scope: Link-Local scope (0x2)]
[Stream index: 195]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
Transaction ID: 0x0000
Flags: 0x8400 Standard query response, No error
Questions: 0
Answer RRs: 1
Authority RRs: 0
0000 33 33 00 00 00 fb 28 a0 6b ce 7c aa 86 dd 60 02 33····(· k·|···`
0010 ff 4c 01 1b 11 01 fe 80 00 00 00 00 00 00 7f b4 ·L·····
0020 d2 bc 07 06 fa c9 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 fb 14 e9 14 e9 01 1b c0 00 00 00
0040 84 00 00 00 00 01 00 00 00 04 10 5f 73 70 6f 74 _spot
0050 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70 ify-conn ect·_tcp
0060 05 6c 6f 63 61 6c 00 00 0c 00 01 00 00 11 94 00 ·local··
0070 3a 1c 53 70 6f 74 69 66 79 20 44 65 73 6b 74 6f :·Spotif y Deskto
0080 70 20 4c 61 75 6e 63 68 65 72 28 32 32 29 10 5f p Launch er(22)·_
0090 73 70 6f 74 69 66 79 2d 63 6f 6e 6e 65 63 74 04 spotify- connect·
00a0 5f 74 63 70 05 6c 6f 63 61 6c 00 1c 53 70 6f 74 _tcp·loc al··Spot
00b0 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61 75 6e ify Desk top Laun
00c0 63 68 65 72 28 32 32 29 10 5f 73 70 6f 74 69 66 cher(22) ·_spotif
00d0 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c y-connec t·_tcp·l
00e0 6f 63 61 6c 00 00 21 00 01 00 00 00 78 00 14 00 ocal··!· ····x···
00f0 00 00 00 da 90 06 73 68 61 69 6e 75 05 6c 6f 63 ·····sh ainu·loc
0100 61 6c 00 c0 6d 00 10 00 01 00 00 11 94 00 0a 09 al··m···
0110 43 50 61 74 68 3d 2f 7a 63 06 73 68 61 69 6e 75 CPath=/z c·shainu
0120 05 6c 6f 63 61 6c 00 00 01 00 01 00 00 00 3c 00 ·local·· ·····<·
0130 04 0a fa 13 54 c0 db 00 1c 00 01 00 00 00 3c 00 ····T··· ·····<·
0140 10 fe 80 00 00 00 00 00 00 7f b4 d2 bc 07 06 fa
0150 c9
Ethernet (eth), 14 bytes
Packets: 148078 · Dropped: 0 (0.0%)
Profile: Default


udp
No. Time Source Destination Protocol Lengt Info
70621 70.811421 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70622 70.811430 10.250.31.171 172.64.41.3 TLSv1… 225 Application Data
70623 70.817255 Intel_ae:c3:ab Broadcast ARP 56 Who has 10.250.16.51? Tell 10.250.13.253
70624 70.818567 AzureWaveTec_86:e6… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.20.163
70625 70.819304 Intel_a8:f6:38 Broadcast ARP 56 Who has 10.250.19.71? Tell 10.250.14.164
70626 70.820276 Intel_f8:a0:2f Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.24.42
70627 70.820278 LiteonTechno_5f:79… Broadcast ARP 56 Who has 10.250.15.79? Tell 10.250.31.201
70628 70.820519 fe80::7fb4:d2bc:70… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70629 70.822004 10.250.19.84 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(22)._spotify-connect._tcp.local SRV 0 0 5595…
70630 70.822279 Intel_20:41:03 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.11.29
70631 70.824004 LiteonTechno_ad:52… Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.15.100
70632 70.824006 Intel_e3:06:11 Broadcast ARP 42 Who has 10.250.13.211? Tell 10.250.15.5
70633 70.824347 Intel_f4:48:13 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.12.162
70634 70.824348 HewlettPacka_52:c1… Broadcast ARP 60 Who has 10.250.19.42? Tell 10.250.1.1
70635 70.825591 Intel_04:00:56 Broadcast ARP 56 Who has 10.250.14.12? Tell 10.250.29.16
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
Transaction ID: 0x0000
Flags: 0x8400 Standard query response, No error
Questions: 0
Answer RRs: 1
Authority RRs: 0
Additional RRs: 4
Answers
Additional records
Ethernet (eth), 14 bytes
Packets: 148078 · Dropped: 0 (0.0%)
Profile: Default


udp
Wireshark · Open Capture File
Look in: /Users/vsameerreddy/Desktop/CN_Skill1_Capture.pcapng
Computer
vsameerred
Name Size Kind Date Modified
CN_Skill1_Capture.pcapng 40....MiB TC...re 11/11/2...1:26 AM
File name: CN_Skill1_Capture.pcapng
Files of type: All Capture Files
Open
Cancel
Help
Automatically detect file type
Format: Wireshark/... - pcapng
Size: 40 MiB, 148,078 data records
Start / elapsed: 2026-11-11 11:23:40 / 00:02:26
Read filter: Apply a read filter ...
Packets: 148078 · Displayed: 96549 (65.2%)
Profile: Default


udp
No. Time Source Destination Protocol Lengt Info
1 0.000000 fe80::a9bd:44b2:1f… ff02::fb MDNS 101 Standard query 0x0000 AAAA BRW00410E848CED.local, "QU" question
2 0.000020 10.250.13.58 224.0.0.251 MDNS 387 Standard query response 0x0000 PTR DESKTOP-OAFQJFL._dosvc._tcp.local SRV 0 0 7680 DESKTOP-OAFQJFL.local …
3 0.000352 10.250.24.112 224.0.0.251 MDNS 297 Standard query response 0x0000 PTR DESKTOP-N6K4O67._dosvc._tcp.local SRV 0 0 7680 DESKTOP-N6K4O67.local …
4 0.000906 10.250.22.154 224.0.0.251 MDNS 651 Standard query response 0x0000 PTR LAPTOP-GNRRAEDD._dosvc._tcp.local SRV 0 0 7680 LAPTOP-GNRRAEDD.local …
5 0.000908 10.250.29.0 224.0.0.251 MDNS 323 Standard query response 0x0000 PTR siri._dosvc._tcp.local SRV 0 0 7680 siri.local TXT A 10.250.29.0 NSEC…
11 0.002461 10.250.19.224 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR DESKTOP-CPH5F2I._dosvc._tcp.local SRV 0 0 7680 DESKTOP-CPH5F2I.local …
13 0.002463 10.250.16.74 224.0.0.251 MDNS 389 Standard query response 0x0000 PTR SREEHARSHA._dosvc._tcp.local SRV 0 0 7680 SREEHARSHA.local TXT A 10.2…
14 0.003179 10.250.15.121 224.0.0.251 MDNS 435 Standard query response 0x0000 PTR AravindPolepeddi._dosvc._tcp.local SRV 0 0 7680 AravindPolepeddi.loca…
15 0.003180 10.250.15.99 224.0.0.251 MDNS 387 Standard query response 0x0000 PTR DESKTOP-4IB1AUF._dosvc._tcp.local SRV 0 0 7680 DESKTOP-4IB1AUF.local …
16 0.003830 10.250.17.185 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR LAPTOP-ON1SJ1LH._dosvc._tcp.local SRV 0 0 7680 LAPTOP-ON1SJ1LH.local …
17 0.004210 10.250.14.169 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 65068…
18 0.004557 10.250.15.109 224.0.0.251 MDNS 719 Standard query response 0x0000 PTR NAGAGANESH._dosvc._tcp.local SRV 0 0 7680 NAGAGANESH.local TXT A 10.2…
19 0.004923 10.250.27.217 224.0.0.251 MDNS 638 Standard query response 0x0000 PTR A3._airplay._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 7000…
20 0.005151 fe80::14dd:5146:9f… ff02::fb MDNS 658 Standard query response 0x0000 PTR A3._airplay._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 7000…
21 0.005833 10.250.29.196 224.0.0.251 MDNS 315 Standard query response 0x0000 PTR Spotify Desktop Launcher(13)._spotify-connect._tcp.local SRV 0 0 4915…
22 0.005835 10.250.14.75 224.0.0.251 MDNS 730 Standard query response 0x0000 PTR LAPTOP-DPKR1QGP._dosvc._tcp.local SRV 0 0 7680 LAPTOP-DPKR1QGP.local …
23 0.006097 10.250.13.35 224.0.0.251 MDNS 291 Standard query response 0x0000 PTR LAPX-0934-890._dosvc._tcp.local SRV 0 0 7680 LAPX-0934-890.local TXT …
Frame 1: Packet, 101 bytes on wire (808 bits), 101 bytes captured (808 bits) on interface en0, id 0
Ethernet II, Src: Intel_a6:86:28 (c4:3d:1a:a6:86:28), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
Destination: IPv6mcast_fb (33:33:00:00:00:fb)
.... ..1. .... = LG bit: Locally administered address (this is NOT the factory d
....1 .... = IG bit: Group address (multicast/broadcast)
Source: Intel_a6:86:28 (c4:3d:1a:a6:86:28)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv6 (0x86dd)
[Stream index: 0]
Internet Protocol Version 6, Src: fe80::a9bd:44b2:1f9:9cf8, Dst: ff02::fb
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0000 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)
.... ..0. .... = Truncated: Message is not truncated
....0 .... = Recursion desired: Don't do query recursively
.... .0.. .... = Z: reserved (0)
....0 .... = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
0000 33 33 00 00 00 fb c4 3d 1a a6 86 28 86 dd 60 0c 33·····= ···(··`·
0010 09 06 00 2f 11 01 fe 80 00 00 00 00 00 00 a9 bd ···/···· ········
0020 44 b2 01 f9 9c f8 ff 02 00 00 00 00 00 00 00 00 D··· ···· ····
0030 00 00 00 00 00 fb 14 e9 14 e9 00 2f aa 25 00 00 ····· ···/·%··
0040 00 00 00 01 00 00 00 00 00 00 0f 42 52 57 30 30 ····· ···BRW00
0050 34 31 30 45 38 34 38 43 45 44 05 6c 6f 63 61 6c 410E848C ED·local
0060 00 00 1c 80 01 ····
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit
Packets: 148078 · Displayed: 96549 (65.2%)
Profile: Default


icmp
No. Time Source Destination Protocol Lengt Info
65505 65.484540 10.250.31.171 8.8.8.8 ICMP 70 Destination unreachable (Port unreachable)
108295 106.138970 10.250.31.171 8.8.8.8 ICMP 70 Destination unreachable (Port unreachable)
114517 112.218697 10.250.31.171 10.250.1.1 ICMP 362 Echo (ping) request id=0xaf0f, seq=0/0, ttl=64 (reply in 114874)
114874 112.486689 10.250.1.1 10.250.31.171 ICMP 362 Echo (ping) reply id=0xaf0f, seq=0/0, ttl=64 (request in 114517)
115820 113.329562 10.250.31.171 10.250.1.1 ICMP 362 Echo (ping) request id=0xaf0f, seq=1/256, ttl=64 (reply in 116128)
116128 113.599186 10.250.1.1 10.250.31.171 ICMP 362 Echo (ping) reply id=0xaf0f, seq=1/256, ttl=64 (request in 115820)
116836 114.315806 10.250.31.171 10.250.1.1 ICMP 362 Echo (ping) request id=0xaf0f, seq=2/512, ttl=64 (reply in 117076)
117076 114.532764 10.250.1.1 10.250.31.171 ICMP 362 Echo (ping) reply id=0xaf0f, seq=2/512, ttl=64 (request in 116836)
117758 115.316741 10.250.31.171 10.250.1.1 ICMP 362 Echo (ping) request id=0xaf0f, seq=3/768, ttl=64 (reply in 118003)
118003 115.519240 10.250.1.1 10.250.31.171 ICMP 362 Echo (ping) reply id=0xaf0f, seq=3/768, ttl=64 (request in 117758)
118609 116.325707 10.250.31.171 10.250.1.1 ICMP 362 Echo (ping) request id=0xaf0f, seq=4/1024, ttl=64 (reply in 118799)
118799 116.464192 10.250.1.1 10.250.31.171 ICMP 362 Echo (ping) reply id=0xaf0f, seq=4/1024, ttl=64 (request in 118609)
Frame 65505: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id
Ethernet II, Src: 5a:4b:6c:28:25:d2 (5a:4b:6c:28:25:d2), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: 5a:4b:6c:28:25:d2 (5a:4b:6c:28:25:d2)
.... ..1. .... = LG bit: Locally administered address (this is NOT the factory d
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 37]
Internet Protocol Version 4, Src: 10.250.31.171, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0x7518 (29976)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0xcaf8 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.31.171
Destination Address: 8.8.8.8
[Stream index: 802]
Internet Control Message Protocol
0000 4c d5 87 52 c1 00 5a 4b 6c 28 25 d2 08 00 45 00 L··R··ZK l(%···E·
0010 00 38 75 18 00 00 40 01 ca f8 0a fa 1f ab 08 08 ·8u···@· ····
0020 08 08 03 03 cc 46 00 00 00 00 45 19 00 84 a7 d4 ····F·· ··E·····
0030 00 00 6d 11 6a c7 08 08 08 08 0a fa 1f ab 00 35 ··m·j··· ·····5
0040 30 11 00 70 00 00 0··p··
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit
Packets: 148078 · Displayed: 12 (0.0%)
Profile: Default

Result / Conclusion

Wireshark was used to capture network packets and inspect Ethernet, IPv4, TCP, and UDP headers. TCP and UDP display filters were applied, packet-header fields were observed, and the header overhead and payload efficiency were calculated. The experiment demonstrates that protocol headers add overhead to transmitted data and affect network efficiency.